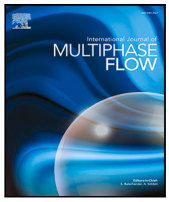
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CNN-based acoustic identification of gas–liquid jet: Evaluation of noise resistance and visual explanation using Grad-CAM

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ABSTRACT

For the analysis of anomalies in a steam generator (SG) of a sodium-cooled fast reactor (SFR), we evaluate the noise resistance of CNN-based acoustic identification methods of gas–liquid two-phase jets and produce visual explanations for their decisions. First, we introduce the water flow sound and the three types of gas–liquid jet sounds, which simulate the background noise and the anomaly sounds, respectively. Second, we produce time–frequency representations for various signal-to-noise ratios (SNRs) and employ AlexNet, VGG16, and ResNet18 to the identification of the gas–liquid two-phase jets. As a result, the best CNN of ResNet18 achieves more than 92% for SNR = 0, −4, −8, and −12 dB and 69% for SNR = −16 and −20 dB. This result indicates that our proposed methods can identify the flow states of gas–liquid two-phase jets in low-level noise environments and detect the gas–liquid two-phase jets even in high-level noise environments. Also, Grad-CAM suggests that ResNet18 focuses on one of the spectrum peaks of the water flow sound and all or part of the signal intensity pattern of the gas–liquid jet sounds. Our proposed methods lead to the safe operation and fast, accurate, and accountable analysis of anomalies in SFR.

1. Introduction

Gas–liquid two-phase flow can be seen in various industries such as chemical processes, nuclear reactors, and power plants (Tan et al., 2021). It is well known that its flow patterns have a strong relationship with heat and mass transfer. Satitchaicharoen and Wongwises (2004) have stated that the flow patterns have an influence on the two-phase pressure drop, holdup, and exchange rate of momentum during the phase-change heat transfer processes. Also, Yao et al. (2019) have noted that some flow patterns may cause various potential problems including fatigue or intensified corrosion of structures. Therefore, it is important and challenging to identify the flow patterns of gas–liquid two-phase flow accurately (Li et al., 2022). From this background, the previous studies have proposed various identification methods of the flow patterns of gas–liquid two-phase flow including differential pressure methods (Shaban and Tavoularis, 2014; Li et al., 2022), optical methods (Peddu et al., 2018; Nie et al., 2022), electrical methods (Paranjape et al., 2012; Li et al., 2023), and acoustic methods (Abbagoni and Yeung, 2016; Zhang et al., 2021a).

Gas–liquid jet, one kind of two-phase flow, can accidentally occur in nuclear reactors including a sodium-cooled fast reactor (SFR). SFR employs liquid sodium as a reactor coolant, which enables the breeding of nuclear fuels while generating electricity. For this property, SFR is expected to improve the efficiency of nuclear fuel utilization and reduce the environmental burden (Kotake et al., 2010). However, the use of sodium carries the risk that the gas–liquid jet with sodium-water reaction is caused when a penetrating crack occurs in a heat transfer tube inside a steam generator (SG) (Hori, 1980). This jet widens the leakage hole (self-wastage phenomenon) and also damages surrounding heat transfer tubes (single/multi-target wastage phenomenon). In some SFR accidents, the gas–liquid jet with sodium-water reaction damaged dozens of heat transfer tubes (Hartwell, 1963; Currie et al., 1990; Alanche et al., 1990; Poplavsky, 1990). However, these serious accidents can be prevented by the early-stage detection. Hence, it is of great importance to develop fast and accurate methods to identify the gas–liquid jet. The identification of gas–liquid jets is also emphasized in other areas including other reactors or piping systems. The literature includes the application of the bubble imaging for an axisymmetric two-phase jet (Akhmetbekov et al., 2010), the identification of gas–liquid dispersion states in a jet bubbling reactor by acoustic signals (Shuai et al., 2020), and the investigation of gas–liquid jet in piping structures by vibration signals (Bamidele et al., 2019). Also, with the breakthrough of computers, more and more studies are introducing

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convolutional neural networks (CNNs) or CNN-based object detection algorithms. CNN is a kind of deep learning (DL) architecture and it usually shows high performance in the image recognition because it is highly invariant to image distortion such as transition, scaling, and tilting. In addition, CNN learns features implicitly while changing the internal parameters by the backpropagation algorithm, so it does not need explicit feature extraction or feature pre-defining (Zhang et al., 2021a). For example, CNN-based optical identification methods have detected overlapping, blurred, and non-spherical bubbles in a wide range of volume gas fractions and calculated the bubble size distribution and the local gas content in a near-to-nozzle region of the bubble jet flow (Poletaev et al., 2020). Also, these methods have recognized the spatiotemporal behaviors of bubbles such as rupturing, merging, and colliding (Wang et al., 2021). As for CNN-based acoustic identification methods, the CNN with a simple structure has achieved an accuracy of 98.74% in the classification of the water flow sound and the three types of gas–liquid jet sounds with slightly different emission hole diameters (Mikami et al., 2023a). Among the various identification methods, with the consideration of the application to SFR, CNN-based acoustic identification methods are gaining more and more attention for the following three reasons. First, the high speed of sound of liquid sodium enables the analysis with a short time response. For example, the speed of sound of liquid sodium exceeds 2000 m/s at about 500 °C (Leibowitz et al., 1971), which is the operation temperature of SFR (Lee et al., 2014). Second, different from optical methods, they are applicable to invisible circumstances including the inside of SG. Third, it can be foreseen that the recognition ability of CNNs makes it possible to identify the flow states of gas–liquid two-phase jets in high-level noise environments: the power of background noise can be 100 times larger than that of anomaly sound in the SFR accidents (Greene et al., 1980). Although our previous study has already investigated the DL-based acoustic identification methods of gas–liquid two-phase jets and the best DL model has reached an accuracy of 99.76% (Mikami et al., 2023a), it has not covered the influence of noise levels. Therefore, in the next step, it is necessary to develop CNN-based acoustic identification methods using time–frequency representations (TFRs) and investigate their noise resistance for various signal-to-noise ratios (SNRs).

Although artificial intelligence (AI) including CNN has achieved high accuracy in most cases, the opacity of black box algorithms has called for research on fairness, accountability, transparency, and explainability (FATE) (Shin, 2020). To satisfy FATE, more and more attention is focused on explainable AI (XAI), which seeks to explain how a black-box AI model produces its outcomes (Moradi and Samwald, 2021). Particularly for CNNs, class activation mapping (CAM) is widely used among various XAI techniques. CAM-based methods produce heatmaps as visual explanations through the linear combination of feature maps. However, CAM requires an additional pooling layer to a target CNN, which limits the interpretation to a specific layer only. For this issue, gradient-weighted CAM (Grad-CAM) has emerged as an alternative method. Grad-CAM can interpret arbitrary layers of a target CNN without any modifications to network architectures, leading to an increase in flexibility (Zhang et al., 2021b). For this property, in the engineering area, Grad-CAM is being introduced to the fault diagnosis of bearings using vibrational signals (Kim and Kim, 2020; Chen and Lee, 2020; Lin and Jhang, 2021). As mentioned before, the gas–liquid jet occurring in SFR accidents may lead to severe accidents. Therefore, its identification methods are required to get high reliability. However, few studies apply Grad-CAM to acoustic identification methods of the flow states of gas–liquid two-phase flows.

Based on the above, the present study aims to evaluate the noise resistance of CNN-based acoustic identification methods of gas–liquid two-phase jets for different SNRs and to produce visual explanations of the decisions of CNNs via Grad-CAM. Our proposed methods can identify the gas–liquid jets with high speed in noisy or invisible environments and explain the basis of their decisions. These things lead to the safe operation and fast, accurate, and accountable anomaly detection or analysis in SFR. This paper begins with a brief description of experimental sound signals of the water flow sound and the three types of gas–liquid jet sounds, which simulate the background noise and anomaly sounds, respectively. This is followed by the explanation of TFRs, CNNs, and Grad-CAM. To reproduce the gas–liquid jet sounds in noisy environments, we add the water flow sound to each gas–liquid jet sound based on SNRs. Then, TFRs are produced based on three time–frequency analysis methods of short-time Fourier transform (STFT), continuous wavelet transform (CWT), and synchrosqueezed wavelet transform (SWT). Also, we employ three typical CNNs of AlexNet, VGG16, and ResNet18, well known as CNNs with high performance. Finally, we discuss the accuracies of our proposed methods in the quadrant classification of the water flow sound and the three types of gas–liquid jet sounds and in the binary classification of the water flow sound and gas–liquid sound. The quadrant classification and the binary classification correspond to the identification of the flow states of gas–liquid two-phase jets and the detection of gas–liquid two-phase jets, respectively. In addition, we produce visual explanations via Grad-CAM and discuss the interest regions for the decisions of CNNs.

2. Experimental data description

The water flow sound and the three types of gas–liquid jet sounds are employed to simulate the background noise and the anomaly sound in SG, respectively. The three types of gas–liquid jet sounds have differences in their emission hole diameters, which covers the enlargement of a leakage hole in the self-wastage phenomenon. We have already referred to all types of sounds in our previous papers (Mikami et al., 2023a,b). Therefore, we just summarize the contents and describe them briefly in this paper.

To measure the water flow sound and the gas–liquid jet sound, we perform two experiments individually. In the experiment on the water flow sound acquisition, a pipe is supported horizontally and water flows inside of it. A hydrophone in a water tank measures the sound signals at a sampling rate of 44 100 Hz for 60 s. In the experiment on the gas–liquid jet sound acquisition, compressed air is released from a nozzle, and the air flow is choked at the nozzle outlet. Three types of nozzles are created by drilling a hole in a 1/4-inch tube, and each nozzle has a nominal emission hole diameter of 0.15, 0.17, and 0.20 mm. The identical hydrophone stated previously is fixed behind the gas–liquid two-phase jet and records the sound signals at a sampling rate of 44 100 Hz for 60 s. In both experiments, we employ the basic air and water as simulant fluids. Then, the measured sound signals are analyzed to investigate the physics of sounds. We produce spectrograms that represent the time variation of spectrums. For the water flow sound, the spectrogram reveals that there are spectrum peaks at around 6000, 10 000, 14 000, and 17 000 Hz. This phenomenon is caused by the coupled waves of cylinder walls and an internal liquid. These peak frequencies can be predicted based on the cylindrical shell theory and the wave equation (Mikami et al., 2023b). As for the gas–liquid jet sounds, all of them have a common signal intensity distribution between 1500 to 5000 Hz, and there is no significant difference in the frequency characteristic between them. According to the bubble acoustics, the detachment, breaking, and coalescing of bubbles emit sound signals into the surroundings. Some studies suggest that the physics of sound signals from bubbles can be described by the Rayleigh-Plesset equation (Zhang et al., 2021a; Mikami et al., 2023a).

3. Methodology

3.1. Time–frequency representations

3.1.1. Time-frequency analysis methods

Time–frequency analysis seeks to decompose a one-dimensional signal along two dimensions: a time axis and a frequency axis (Gardner

and Magnasco, 2006). TFRs obtained from time–frequency analysis have information on time, frequency, and signal intensity. Therefore, TFRs highlighting signal features are often used as inputs of CNNs. In the present study, in addition to the basic time–frequency analysis methods of STFT and CWT, we generate TFRs using SWT, which is effective for acquiring higher time and frequency resolutions (Zhang et al., 2021a). Each method is described in the following.

First, given a signal $s(t)$, STFT $F_s(\omega, t)$ is defined as (Najmi and Sadowsky, 1997):

$$F_s(\omega, t) = \int s(\tau) w(\tau - t) e^{-i\omega\tau} d\tau, \tag{1}$$

where t is a time, ω is an angular frequency, respectively, and i represents an imaginary unit. Also, $w(t)$ is a window function, and the hanning window is employed in the present study. In STFT, the signal is divided into small sequential or overlapping data frames, and the fast Fourier transform is applied to each windowed data frame (Kıymık et al., 2005). Through this process, STFT assumes that a generic non-stationary signal can be considered stationary. STFT has fixed time and frequency resolutions over the entire time–frequency space once the window size is chosen. Also, according to the principle of uncertainty, increasing the time resolution decreases the frequency resolution and vice versa. Although the fixed resolutions enable an easier interpretation in terms of harmonics, they make it difficult to analyze of signals with fast transitions (Gu and Bollen, 2000; Han et al., 2014).

Second, CWT $W_s(a, b)$ is defined as follows (Najmi and Sadowsky, 1997):

$$W_s(a, b) = \frac{1}{\sqrt{a}} \int s(t) \overline{\psi\left(\frac{t-b}{a}\right)} dt, \tag{2}$$

where a and b represent a scaling parameter and a time shift parameter, respectively, and the parameter $1/\sqrt{a}$ ensures energy preservation (Lin and Qu, 2000). $\overline{\psi(t)}$ is the complex conjugate of a mother wavelet $\psi(t)$. CWT is performed by the projection of a signal onto wavelets obtained from the mother wavelet by transitions and dilations, and the calculation of correlation values of the signal and wavelets (Hardalaç et al., 2007). CWT provides high frequency resolutions at low frequencies and high time resolutions at high frequencies (Najmi and Sadowsky, 1997); CWT is more effective to estimate frequency locations at lower frequencies and time locations at higher frequencies. Also, we employ the complex Morlet wavelet as the mother wavelet $\psi(t)$. As shown in Eq. (2), CWT is a function of the scaling parameter a and the time shift parameter b, therefore it can be called a time-scale analysis. However, CWT is more useful if transformed into a time–frequency analysis like STFT. The transformation from the scale a into the pseudo-frequency f_{pseudo} is calculated by (Rahaman and Wheeler, 2005; Łuczak, 2023):

$$f_{\text{pseudo}} = \frac{f_c f_s}{a}, \tag{3}$$

where f_s is a sampling rate and f_c is a center frequency of the complex Morlet mother wavelet. CWT results shown below are represented in the time–frequency region through this process.

To motivate the idea of SWT, let us start with a purely harmonic signal with an amplitude A and an angular frequency ω:

$$s(t) = A\cos(\omega t). \tag{4}$$

Also, assuming that the Fourier transform of a mother wavelet $\hat{\psi}(\xi)$ is concentrated on the positive frequency axis: $\hat{\psi}(\xi) = 0$ for $\xi < 0$, CWT $W_s(a, b)$ in Eq. (2) can be rewritten as:

$$\begin{aligned} W_s(a, b) &= \frac{1}{2\pi} \int \hat{s}(\xi) a^{1/2} \overline{\hat{\psi}(a\xi)} e^{ib\xi} d\xi \\ &= \frac{A}{4\pi} \int [\delta(\xi - \omega) + \delta(\xi + \omega)] a^{1/2} \overline{\hat{\psi}(a\xi)} e^{ib\xi} d\xi \\ &= \frac{A}{4\pi} a^{1/2} \overline{\hat{\psi}(a\omega)} e^{ib\omega}, \end{aligned} \tag{5}$$

where $\hat{s}(\xi)$ is the Fourier transform of the signal $s(t)$, $\overline{\hat{\psi}(a\xi)}$ is the complex conjugate of $\hat{\psi}(a\xi)$, and δ denotes the delta function. If $\hat{\psi}(\xi)$ has signal intensity around $\xi = \omega_0$, CWT $W_s(a, b)$ is spread out over a region around $a = \omega_0/\omega$. This leads to the suggestion to compute a candidate instantaneous frequency $\omega_s(a, b)$ and it is calculated from Eq. (5):

$$\omega_s(a, b) = -i \left(W_s(a, b)\right)^{-1} \frac{\partial}{\partial b} W_s(a, b). \tag{6}$$

In the next step of synchrosqueezing, the information on the time-scale plane is transferred into the time–frequency plane, in which every point (a, b) is converted to another point $(\omega_s(a, b), b)$. CWT $W_s(a, b)$ is computed only at discrete values a_k, and SWT $T_s(\omega_l, b)$ is calculated only at centers ω_l of the frequency range $[\omega_l - \Delta\omega/2, \omega_l + \Delta\omega/2]$ as follows (Daubechies et al., 2011):

$$T_s(\omega_l, b) = \frac{1}{\Delta\omega} \sum_{a_k : |\omega(a_k, b) - \omega_l| \le \Delta\omega/2} W_s(a_k, b) a_k^{-3/2} (\Delta a)_k, \tag{7}$$

where $a_k - a_{k-1} = (\Delta a)_k$ and $\omega_l - \omega_{l-1} = \Delta\omega$. The synchrosqueezing operation enables the enhancement of the frequency localization through the reallocation of the energy of wavelet coefficients (Amezquita-Sanchez and Adeli, 2015).

3.1.2. Time–frequency representations with different signal-to-noise ratios

Each sound signal is divided into 600 segments. From each segment, we produce the TFRs using STFT, CWT, and SWT. All TFRs in the present study have time information for 0.1 s and frequency information between 0 to 44 100/2 Hz, and they are normalized to be used as the DL data. Also, to reproduce the gas–liquid jet sound in noisy environments, the water flow sound is added to each gas–liquid jet sound according to each SNR, and the SNR is defined as:

$$\text{SNR} = 10 \log_{10} \left(\frac{P_{\text{signal}}}{P_{\text{noise}}} \right), \tag{8}$$

where P_{signal} and P_{noise} represent the powers of a signal and noise, respectively. The present study introduces the cases of SNR = $0, -4, -8, -12, -16,$ and -20 dB. Fig. 1 shows the example TFRs of the gas–liquid jet sound with the addition of the water flow sound for different SNRs. In this figure, to give representative examples, we use the gas–liquid jet sound from a emission hole diameter of 0.15 mm. First, we compare the difference between the signal intensity distribution of STFT, CWT, and SWT. In STFT, the signal intensity appears similarly in all frequency bands because STFT has a fixed frequency resolution as stated previously. On the other hand, the signal intensity of CWT shows a frequency-wise band-like pattern. This is caused by the property of CWT that the signal intensity spreads out over a region around the center frequency of the scaled complex Morlet wavelet. In SWT, the synchrosqueezing operation recalculates CWT coefficients, therefore, the signal intensity is localized. Second, we discuss the signal intensity distribution of the water flow sound and the gas–liquid jet sound for different SNRs. For SNR = 0 dB (most left column), the signal intensity of the gas–liquid jet sound between 1500 to 5000 Hz appears with a similar degree as that of the water flow sound with several spectrum peaks. As lower SNRs, that is higher-level noises, the signal intensity of the gas–liquid jet sound attenuates while that of the water flow sound becomes dominant. For SNR = -16 and -20 dB, the signal intensity of the gas–liquid jet sound is not observed clearly. This motivates the present study to introduce the CNNs with high recognition ability for the acoustic identification of the gas–liquid jet.

3.2. Convolutional neural networks

CNN is a well-known DL architecture inspired by the natural visual perception mechanism of living creatures (Gu et al., 2018). CNN mainly consists of an input layer, convolutional layers, pooling layers, fully connected layers, and an output layer. Convolutional layers extract features of input data through the convolution of input data and kernels, also called filters, and the calculation by an activation function.

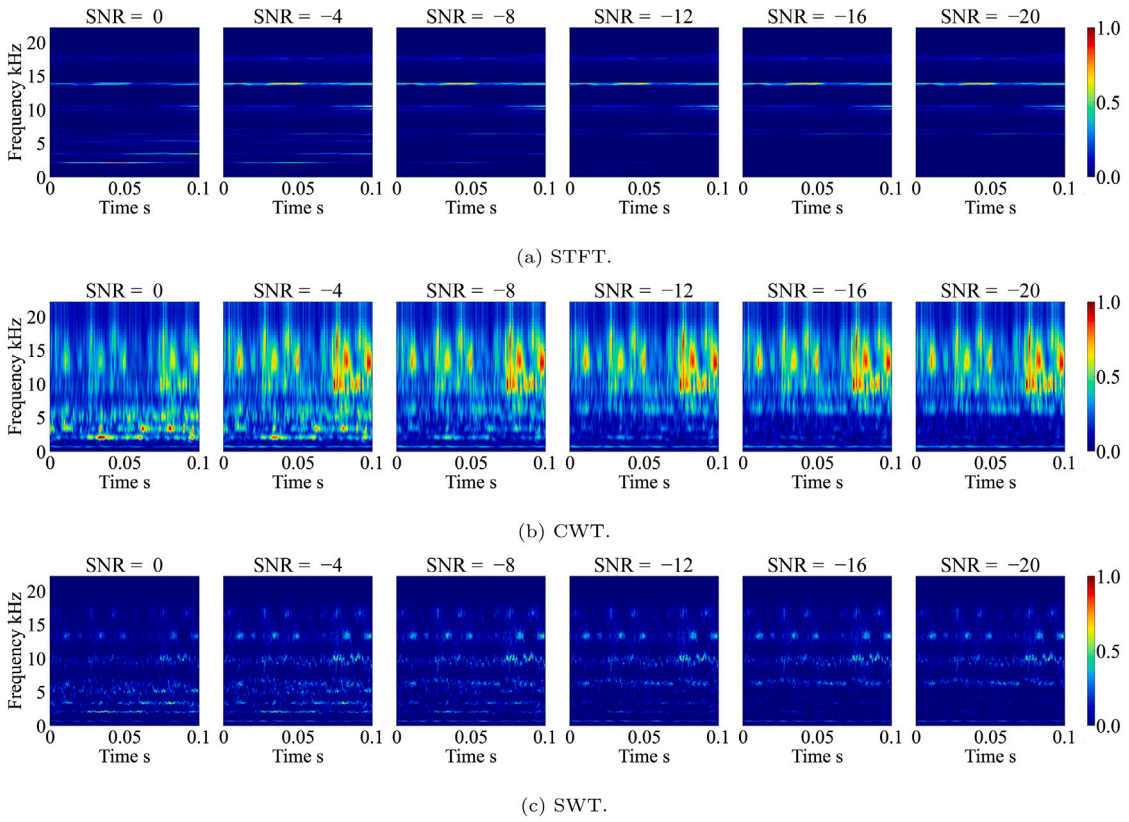


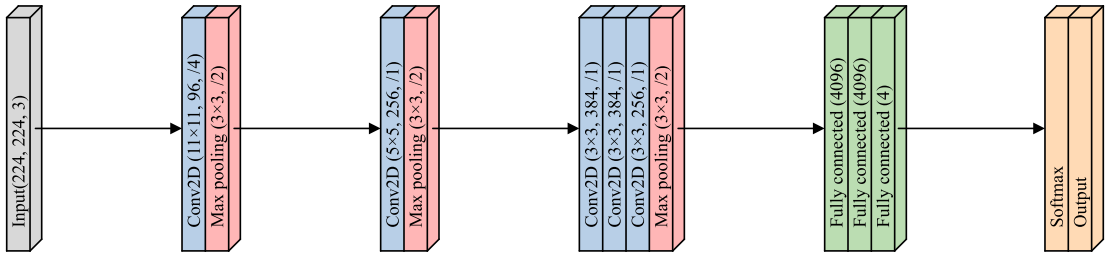
Fig. 1. TFRs of the gas–liquid jet sound (signal) with the addition of the water flow sound (noise) for different SNRs. Each from left to right shows the TFR for SNR = $0, -4, -8, -12, -16,$ and -20 dB.

The shallower convolutional layers usually capture low-level features such as edges, lines, and corners, while the deeper convolutional layers learn high-level features including structures, objects, and shapes by combining low-level ones (Han et al., 2018). Following convolutional layers, pooling layers reduce the dimension of the data by sliding a window to apply selected operations such as maximizing or averaging. Pooling layers aim to create invariance to small shifts and distortions. After convolutional and pooling layers, the data are transformed into a one-dimensional vector, which is the input data of fully connected layers. Fully connected layers consist of one or more layers in the last few layers of CNNs to summarize the information conveyed by lower-level layers. Current CNNs use the rectified linear unit (ReLU) function, performing best in most cases, as an activation function in convolutional layers and fully connected layers (Sharma et al., 2020). In the last fully connected layer, the softmax function is usually employed to compute the probability in classification problems (Gkioxari et al., 2015; Sarıgül et al., 2019).

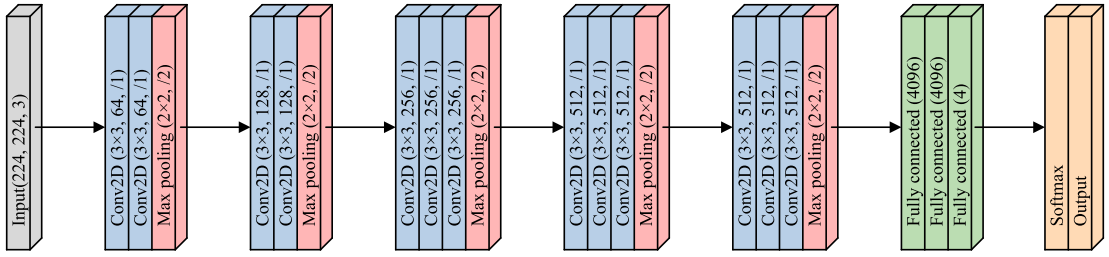
In the identification of the gas–liquid jet, we introduce typical CNNs of AlexNet (Krizhevsky et al., 2017), VGG16 (Simonyan and Zisserman, 2014), and ResNet18 (He et al., 2016) with high performance in ImageNet Large Scale Visual Recognition Challenge (ILSVRC). All CNNs have four neurons in the final fully connected layers to classify the four types of sound of the water flow sound and the three types of gas–liquid jet sounds from emission hole diameters of 0.15, 0.17, and 0.20 mm. Also, to make CNNs more adaptable to the identification of gas–liquid jet, the present study only refers to architectures of AlexNet, VGG16, and ResNet18, and all trainable parameters are initialized randomly and updated by the backpropagation. Fig. 2 shows the architectures of AlexNet, VGG16, and ResNet18. AlexNet is the winner of ILSVRC 2012 and has made a breakthrough in the development of CNN architectures. AlexNet consists of five convolutional layers, three pooling layers, and three fully connected layers. The main characteristics of AlexNet are the ReLU function and dropout. ReLU function effectively improves the training speed and prevents the problem of gradient disappearance and gradient explosion. Therefore, the ReLU function makes it easy to train deeper networks (Nair and Hinton, 2010). Also, to reduce the degree of overfitting, dropout stops the update of neurons with a certain probability. Thus the dependence on local neurons is reduced and the generalization ability of the model is improved (Hinton et al., 2012). These techniques have become standard in CNN architectures. VGG16 is the second in ILSVRC 2014 and consists of thirteen convolutional layers, five pooling layers, and three fully connected layers. VGG16 continues the characteristics of classical simple structure, which can be confirmed in the flexible 3×3 convolutional layers stacked in depth. The increased depth and the smaller kernels can decrease the overall parameters, which promotes the fitting capacity (Hong et al., 2021). However, VGG16 has some drawbacks in the application. One of them is that VGG16 lacks an effective method to prevent the disappearance of gradients. This can cause the problems such as slow convergence speed, degradation problem, and gradient explosion in the training session (Yang et al., 2021). ResNet is the winner of ILSVRC 2015 and consists of seventeen convolutional layers, two pooling layers, and one fully connected layer. The main component of ResNet is a residual block with convolutional layers and a shortcut connection. In each residual block, an input value or a linear projection of it is added to the value processed through convolutional layers, and this value is an output value. The residual learning improves the degradation problem of higher training losses with deeper CNNs (Fang et al., 2021). Therefore, it enables the effective learning of deep CNNs.

3.3. Gradient-weighted class activation mapping

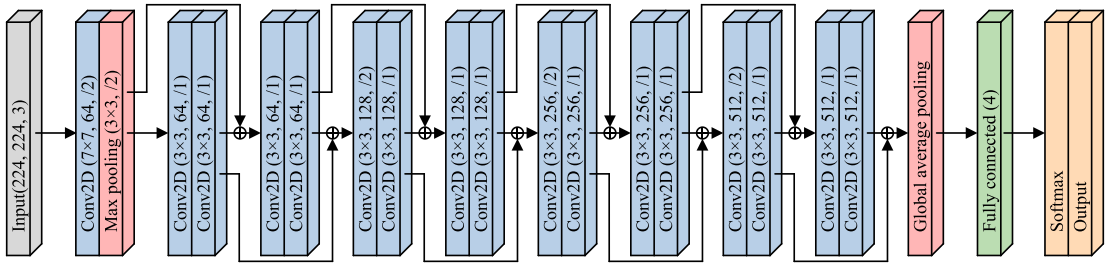
Grad-CAM (Selvaraju et al., 2017) is a technique to produce visual explanations for decisions from CNNs. It uses the gradients flowing into the final convolutional layer to understand the importance of each neuron for a decision of interest. Through these gradients, Grad-CAM



(a) AlexNet.



(b) VGG16.



(c) ResNet18.

Fig. 2. Architectures of CNNs introduced in the present study. Input layers have a shape with 224 pixels in width and height, and three RGB channels. Conv2D represents a two-dimensional convolutional layer and its parameters are denoted as "(kernel size, filter, /stride)". Max pooling parameters are denoted as "(window size, /stride)". A fully connected layer parameter is denoted as "(neuron)".

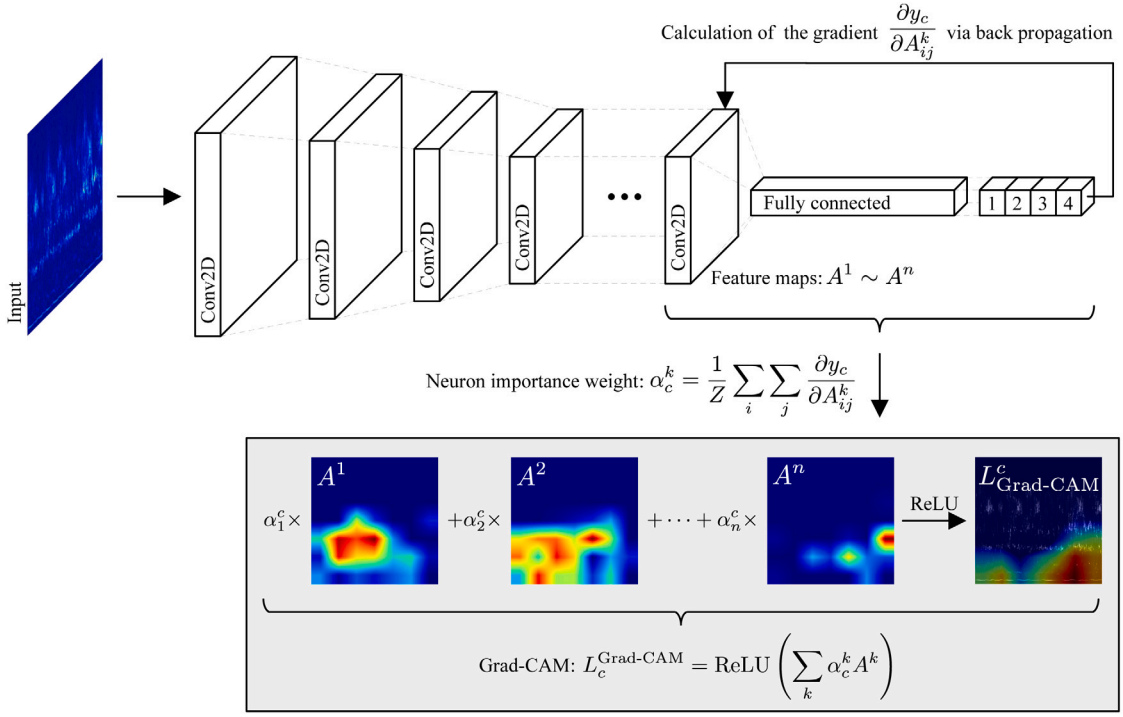


Fig. 3. An overview of Grad-CAM algorithm.

Table 1
Dataset description for all SNRs. Shape parameters are denoted as "(height, width, color channel)". Data amount parameters are denoted as "(training data, validation data, test data)".

TFR	Shape	Data amount			
	(pixel, pixel, channel)	Class W	Class J0.15	Class J0.17	Class J0.20
STFT CWT SWT	(224, 224, 3)	(450, 100, 50)	(450, 100, 50)	(450, 100, 50)	(450, 100, 50)

Table 2
Adjusted hyperparameters of AlexNet, VGG16, and ResNet18.

CNN model	Batch size	Epoch	Learning rate	Optimizer	Loss function
AlexNet	5	100	1.0×10^{-6}	Adam	Cross entropy loss function
VGG16			5.0×10^{-7}		
ResNet18			1.0×10^{-5}		

produces a localization map, which highlights the important regions in the prediction of an input image. Fig. 3 illustrates an overview of the Grad-CAM algorithm. First, in the forward calculation for an input image, we compute an output value of a CNN for class c represented by y_c and kth output of the final convolutional layer called a feature map represented by A^k. Second, we calculate the gradient of y_c with respect to A^k through the backpropagation. Using this gradient, the neuron importance weight α_c^k is obtained as follows:

$$\alpha_c^k = \frac{1}{Z} \sum_i \sum_j \frac{\partial y_c}{\partial A_{ij}^k}, \tag{9}$$

where Z is the number of the components in each A^k. Eq. (9) means the global average pooling of the gradient of the output value with respect to the feature map. The weight α_c^k represents a partial linearization of CNNs and captures the importance of the kth feature map for the target class c. In the next step, the class-discriminative localization map $L_c^{\text{Grad-CAM}}$ is calculated through a weighted combination of forward feature maps and the ReLU function:

$$L_c^{\text{Grad-CAM}} = \text{ReLU}\left(\sum_k \alpha_c^k A^k \right). \tag{10}$$

Notice that the size of $L_c^{\text{Grad-CAM}}$ is the same as the feature map in the final convolutional layer. Also, the ReLU function is applied to the linear combination of maps to focus on a positive influence on the class of interest: the intensity of each pixel should be increased to get larger y_c.

4. Results and discussions

4.1. Dataset and hyperparameters

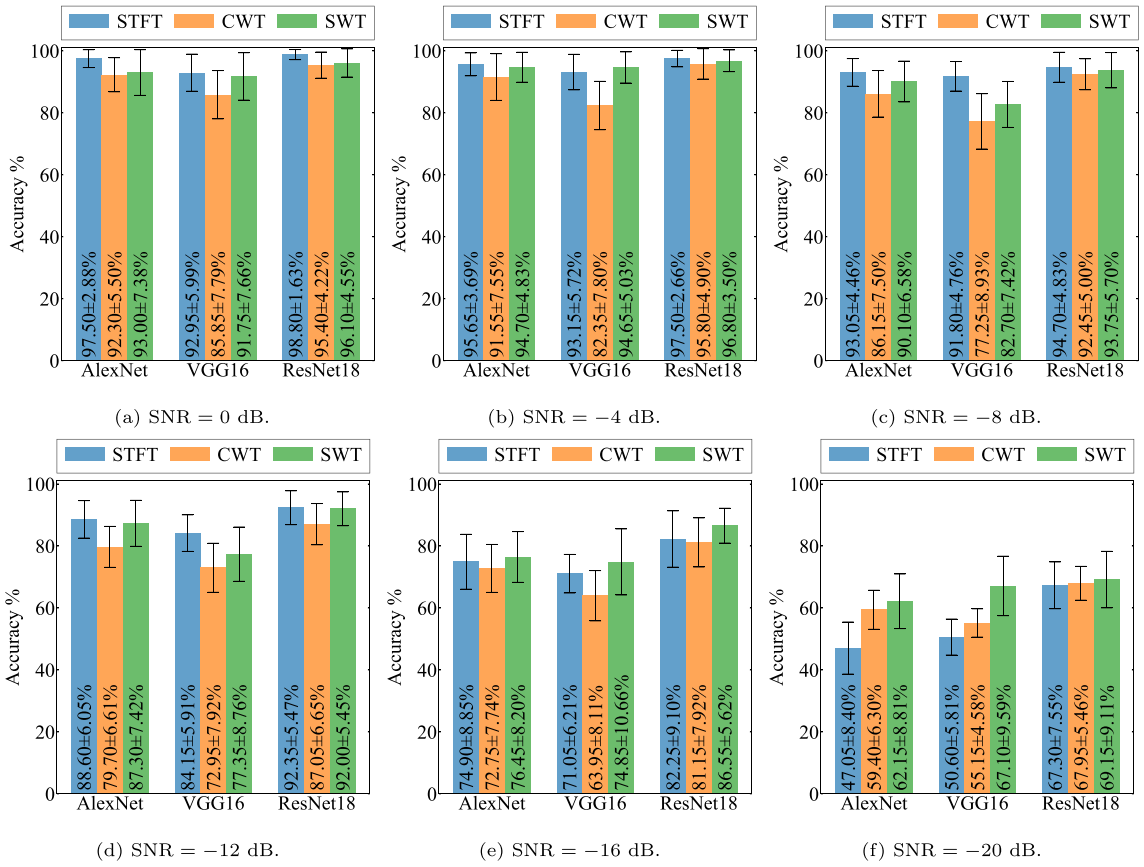
TFRs of the water flow sound and the three types of gas–liquid jet sounds are used as inputs of CNNs. Hereafter, the class of the water flow sound is called as Class W and the classes of the three types of gas–liquid jet sounds from emission hole diameters of 0.15, 0.17, and 0.20 mm are referred to as Class J0.15, J0.17, and J0.20, respectively. As mentioned in Section 3, TFRs are three types of STFT, CWT, and SWT, and the water flow sound is added to each gas–liquid jet sound according to SNR $= 0, -4, -8, -12, -16,$ and -20 dB to reproduce the gas–liquid jet sound in noisy environments.

Table 1 shows the detailed dataset description. The shape of each TFR is 224 pixels in width and height, and three RGB channels. In each of STFT, CWT, and SWT, 600 TFRs are generated from each class, of which 450 are used for training data, 100 for validation data, and 50 for test data. Also, the present study employs K-fold cross-validation to evaluate the generalization performance of our proposed methods (Refaeilzadeh et al., 2016). In K-fold cross-validation, all data is partitioned into K subsets of equal size. $K-1$ subsets are used as the training data, and the remaining subset is used as the test data. This procedure is repeated until each of the K subsets has served as the test data. The average performance in K trials is the cross-validated performance (Berrar, 2019). We set the value of K to ten, therefore, the ratio of the amounts of the training data and test data is $9:1$. Also, the amount of TFRs of each class is the same in all subsets. In the present study, we firstly adjust hyperparameters using the fixed validation data and repeat the training and test of CNNs ten times.

The adjusted hyperparameters are shown in Table 2. The batch size, epoch, and learning rate are determined based on the history of accuracy and loss for SNR = 0 dB. Among them, the epoch and learning rate have an influence on the convergence of the accuracy and loss, which may change the performance of the trained CNNs. Therefore, they are characteristic hyperparameters in our CNNs. We confirm that the accuracy and loss are well converged using the epoch and learning rate in Table 2. The readers may refer to them in Appendix. Also, we employ the Adam optimizer as an optimization algorithm (Kingma and Ba, 2014) and the cross-entropy loss function as a loss function. The Adam optimizer can effectively adjust a learning rate in the training session and it is widely used in the optimization of DL models. The cross-entropy loss function is a standard choice in using DL models with the softmax function in output layers because a differential value of the function can be smoothly calculated in the backpropagation. To investigate the robustness of our proposed methods, we also use the same hyperparameters shown in Table 2 for other SNRs.

4.2. Evaluation of noise resistance

AlexNet, VGG16, and ResNet18 are constructed, trained, validated, and tested with the TensorFlow framework. Fig. 4 shows the accuracies of AlexNet, VGG16, and ResNet18 combined with STFT, CWT, and SWT. These are the averages in K-fold cross-validation and error bars represent the standard deviations. As shown in Fig. 4(a), ResNet18 with STFT classifies the water flow sound and the three types of gas–liquid jet sounds with high accuracy of 98.80% for SNR $= 0$ dB where the power of the gas–liquid jet sound is equal to that of the water flow sound. The ability to distinguish between the water flow sound and the gas–liquid jet sound is reasonable since the characteristic frequency bands are different between the gas–liquid jet sound and the water flow sound, that is the former and latter exist around 1500 to 5000 Hz and around 6000 Hz or more, respectively. In addition to this, the recognition ability of CNNs enables the identification of three types of gas–liquid jet sounds with slight differences. On the other hand, the accuracies of all methods monotonically decrease as lower SNRs, that is, higher-level noises. This is because the signal intensity of the gas–liquid jet sound is more buried, which reduces the difference between the TFRs of the four types of sounds (see Fig. 1). Nevertheless, the recognition ability of CNNs keeps a certain level of accuracy. Particularly for SNR $= -20$ dB, a relatively high-level noise environment where the power of the gas–liquid jet sound is 100 times smaller than that of the water flow sound, ResNet18 with SWT reaches



(a) SNR = 0 dB. (b) SNR = −4 dB. (c) SNR = −8 dB.

(d) SNR = −12 dB. (e) SNR = −16 dB. (f) SNR = −20 dB.

Fig. 4. Average accuracies of AlexNet, VGG16, and ResNet18 with STFT, CWT, and SWT in K-fold cross-validation. In average accuracies of each CNN, each from left to right represents the cases of STFT (shown in blue), CWT (shown in orange), and SWT (shown in green). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

an accuracy of 69.15% in the classification of the four types of sounds. Also, we calculate an accuracy in the binary classification of the water flow sound and the gas–liquid jet sound and it becomes 87.85%. These results indicate that CNNs work effectively in the acoustic identification of the gas–liquid jets in noisy environments.

Fig. 5 shows the classification results of the best methods for each SNR as confusion matrixes. In these confusion matrixes, horizontal and vertical directions represent the true and predicted classes, respectively. Also, the (m, n) elements of the confusion matrixes where $m, n = 2, 3, 4$ are surrounded by the blue square to clarify the distinction between the water flow sound and the gas–liquid jet sound. From Fig. 5, although ResNet18 accurately identifies four types of sounds in low-level noise environments, the number of misclassified data (non-diagonal elements) increases in higher-level noise environments. Also, many of the misclassified data are produced in the classification of three types of the gas–liquid jet sounds as shown in the classification results inside the blue square. The reason for this is that the difference between the signal intensity distribution of the three types of gas–liquid jet sounds is decreased through the normalization by the dominant signal intensity of the water flow sound. On the other hand, the misclassified data in the binary classification between the water flow sound and the gas–liquid jet sound is not significant as shown in the classification results outside the blue square. These results suggest that our proposed method can correctly identify the flow states of the gas–liquid two-phase jet in low-level noise environments and detect the gas–liquid two-phase jets in high-level noise environments.

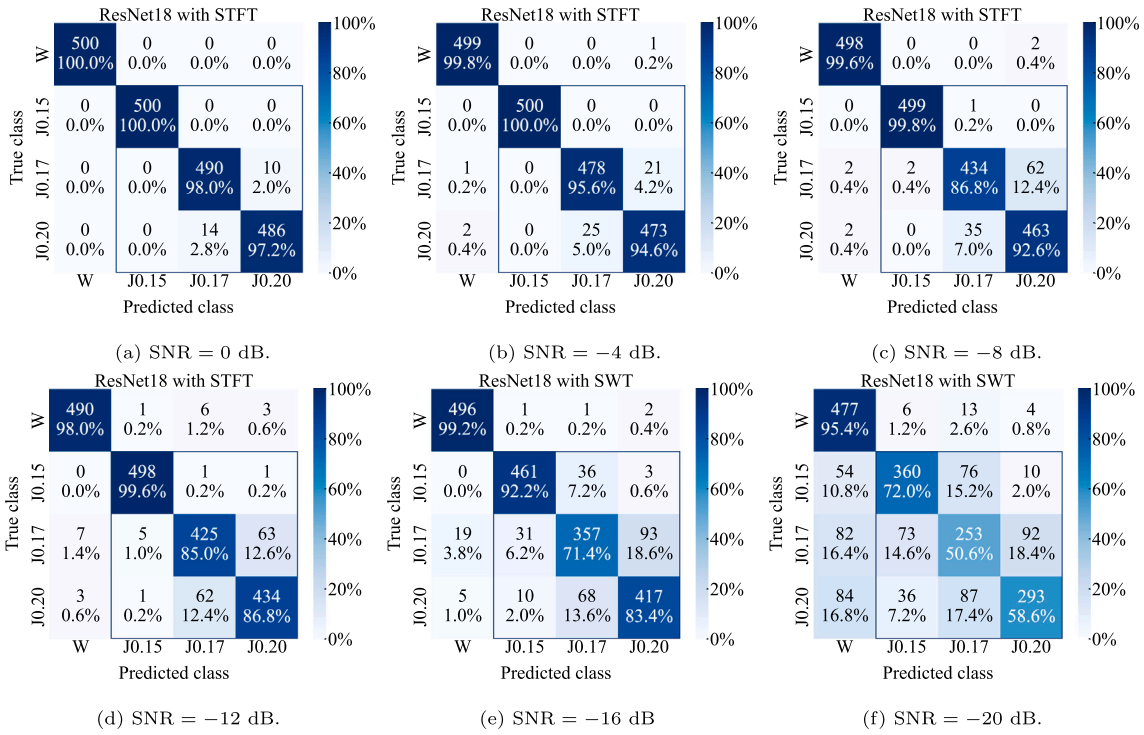
The best CNN with the highest accuracy is ResNet18 for all SNRs. This result is valid since AlexNet, VGG16, and ResNet18 are developed in this order, and ResNet18 records the smallest loss among them in ILSVRC. Also, the best TFR is STFT for SNR $= 0, -4, -8,$ and -12 dB, while it is SWT for SNR $= -16$ and -20 dB. We take two reasons for the effectiveness of STFT. First, we speculate that the acoustic signals have periodicity, for which STFT with periodic basis functions is suitable. Second, TFRs have broad signal intensity distribution in the present study: the water flow sound is in the high-frequency band, and the gas–liquid jet sounds are in the low-frequency band. Therefore, STFT with a fixed frequency resolution may produce superior signal intensity characteristics for all sound types. On the other hand, CWT is inferior to STFT and SWT for most SNRs. This is caused by the band-like signal intensity based on the CWT property. Next, we focus on STFT and SWT with high accuracy for most SNRs and discuss the advantage of SWT for SNR $= -16$ and -20 dB. In such high-level noise environments, the important thing is that TFRs represent the buried signal intensity of the gas–liquid jet sound with a large degree. To quantify the signal intensity of the gas–liquid jet sound buried in noise-added TFRs, we define the relative signal intensity of the gas–liquid jet sounds for STFT:

$$F_s'(\omega, t)_{\text{Class J}} = |F_s(\omega, t)_{\text{Class J}} - F_s(\omega, t)_{\text{Class W}}|, \tag{11}$$

where the subscripts represent the sound types, and Class J corresponds to either Class J0.15, Class J0.17, or Class J0.20 (Class J ∋ Class J0.15, Class J0.17, Class J0.20). We also define the relative signal intensity of the gas–liquid jet sounds for SWT:

$$T_s'(\omega_l, b)_{\text{Class J}} = |T_s(\omega_l, b)_{\text{Class J}} - T_s(\omega_l, b)_{\text{Class W}}|, \tag{12}$$

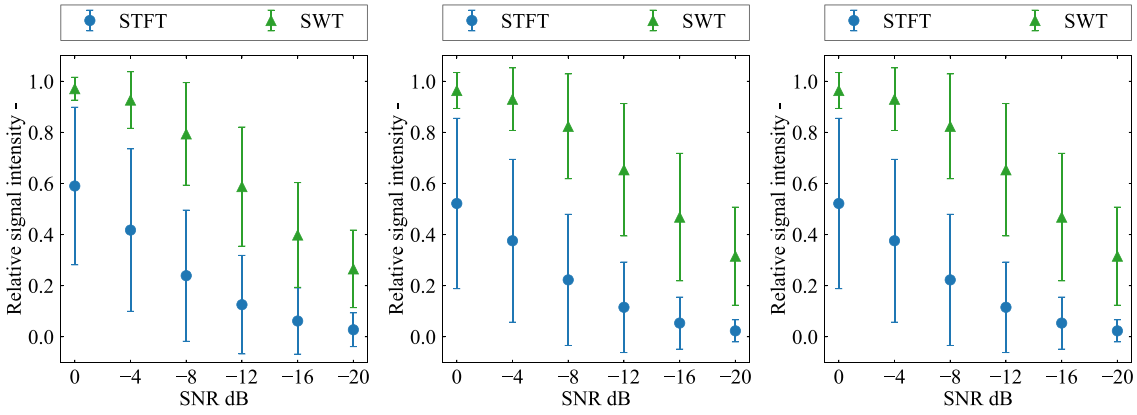
where the notation of the subscripts is the same as above. From all TFRs based on STFT and SWT, we obtain 600 values of $F_s'(\omega, t)_{\text{Class J}}$ and $T_s'(\omega_l, b)_{\text{Class J}}$ for each SNR. Then, we calculate the maximum value of each $F_s'(\omega, t)_{\text{Class J}}$ and $T_s'(\omega_l, b)_{\text{Class J}}$ between 1500 to 5000 Hz, a characteristic frequency band of the gas–liquid jet sound, to extract the most significant signal intensity of the buried gas–liquid jet sound.



(a) SNR = 0 dB. (b) SNR = −4 dB. (c) SNR = −8 dB.

(d) SNR = −12 dB. (e) SNR = −16 dB (f) SNR = −20 dB.

Fig. 5. Confusion matrixes of the best method for each SNR. Horizontal and vertical directions represent true and predicted classes, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



(a) Class J0.15. (b) Class J0.17. (c) Class J0.20.

Fig. 6. Maximum value of the relative signal intensity of the gas–liquid jet sound in noise-added TFRs. Blue circle and green triangle denote the average values of the maximum relative signal intensity of STFT and SWT, respectively.

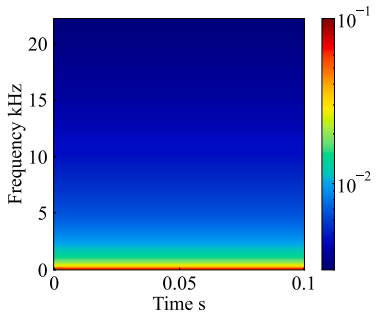


Fig. 7. Distribution of the product of $a_k^{-3/2}$ and $(\Delta a)_k$ in the definition of SWT.

Fig. 6 illustrates the average value of the maximum relative signal intensity. In all classes, both relative signal intensities of STFT and SWT decrease as lower SNRs, and the relative signal intensity of SWT is consistently larger than that of STFT. Particularly for SNR = −20 dB, the value of SWT is about seven to nine times larger than that of STFT. This result is caused by a property of SWT. The present study sets the angular frequency increment $\Delta\omega$ in Eq. (7) to a constant value, which makes the scale increment an inconstant value. Therefore, the product of $a_k^{-3/2}$ and $(\Delta a)_k$ may affect signal intensity distributions. Fig. 7 shows the distribution of $a_k^{-3/2}(\Delta a)_k$ in the present study. It can be seen from this distribution that $a_k^{-3/2}(\Delta a)_k$ increases as lower frequencies, and this increase is outstanding particularly below 5000 Hz. This is because the increase of $(\Delta a)_k$ is significant while $a_k^{-3/2}$ decreases as lower frequencies. From the above, the property of SWT intensifies the signal intensity of the gas–liquid jet sounds with a characteristic

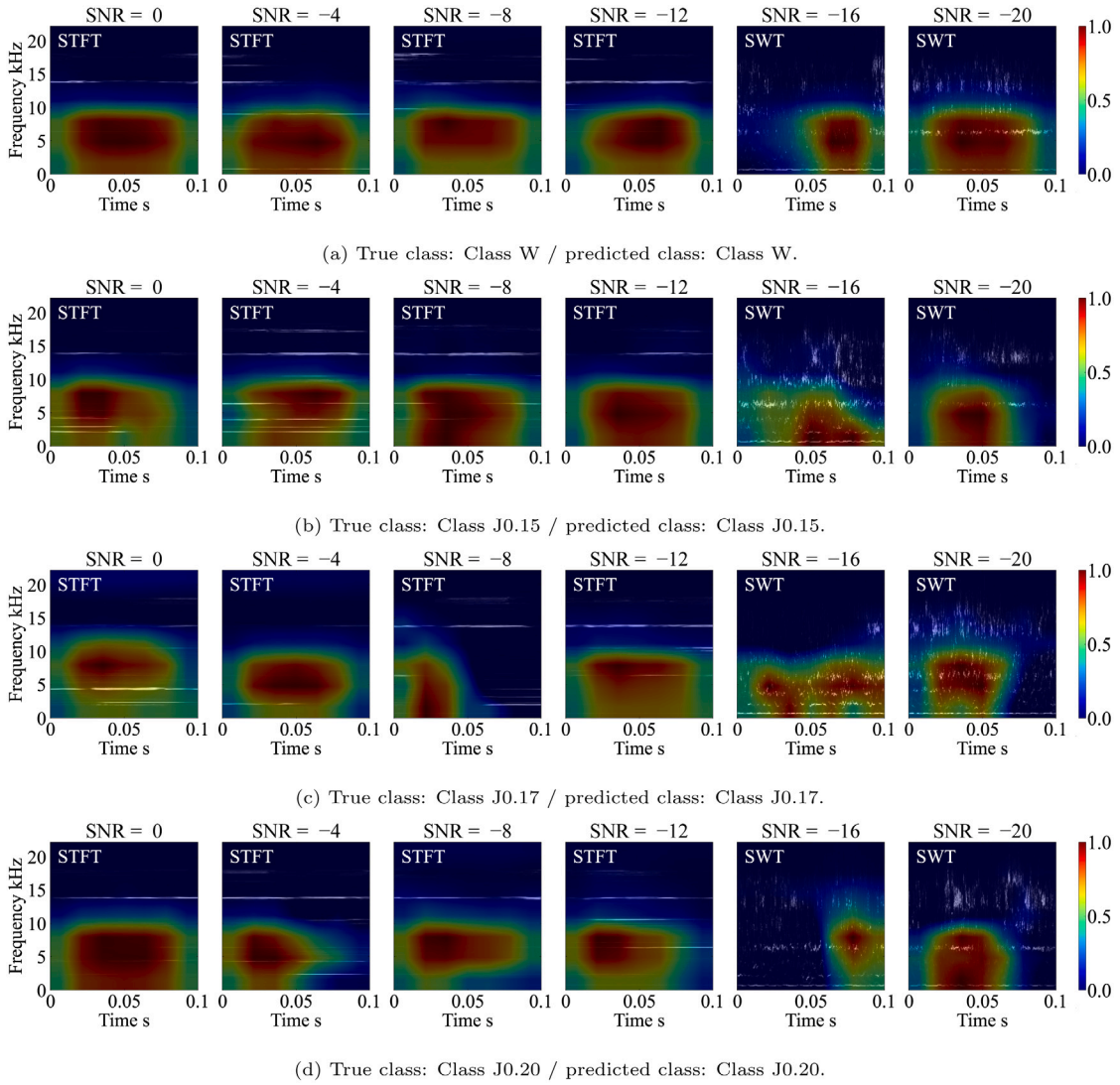


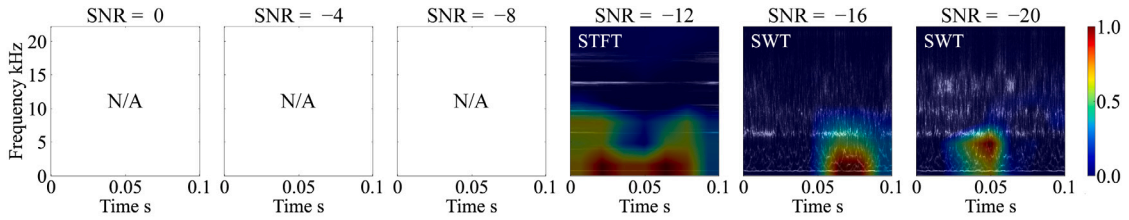
Fig. 8. Visual explanations of correct decisions made by ResNet18 for each SNR. Input images in gray-scale are overlapped with localization maps in the RGB scale obtained by Grad-CAM. Each from left to right shows the visual explanation for SNR = 0, −4, −8, −12, −16 and − 20 dB. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

frequency band between 1500 to 5000 Hz. Therefore, the performance of SWT exceeds that of STFT in high-level noise environments such as SNR = −16 and − 20 dB.

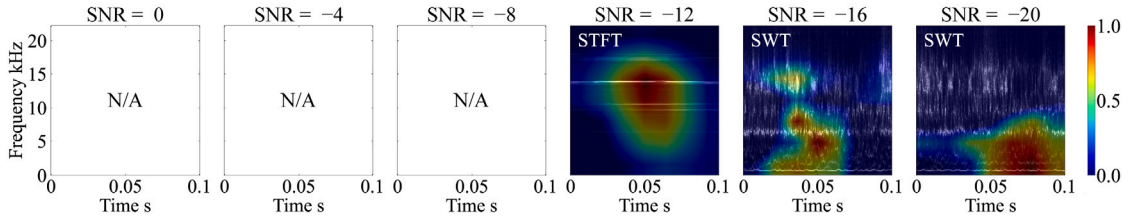
4.3. Visual explanation using Grad-CAM

We apply Grad-CAM to the best methods for each SNR. In the present study, although the localization maps have seven values in both width and height, they are resized to the same size as the input images with the bi-linear interpolation to clarify the important regions for the decisions of CNNs. Fig. 8 shows the visual explanations of correct decisions made by ResNet18 in the classification of four types of sounds. In this figure, the color images are the localization maps obtained by Grad-CAM, and the black and white images are the TFR images converted from the RGB scale to the gray scale. The TFR images are overlapped with the localization maps, from which one can interpret the interest region of ResNet18 for the TFR images. For all sound types and SNRs, the localization map of Grad-CAM is located below about 10 000 Hz. This region may be an important region because part of the signal intensity of the water flow sound and all of the signal intensity of the gas–liquid jet sounds exist in this range. This result suggests that ResNet18 identifies the water flow sound based on its spectrum peak around 6000 Hz and the gas–liquid jet sounds based on all or part of their signal intensity pattern between 1500 to 5000 Hz.

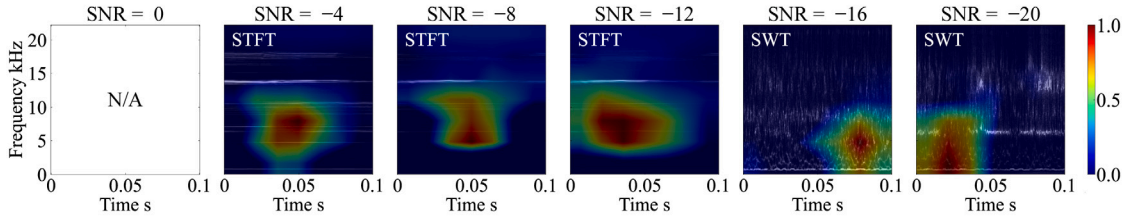
Next, we focus on the visual explanation of incorrect decisions made by ResNet18 in the binary classification of the water flow sound and the gas–liquid jet sound. Fig. 9 is the visual explanation where the true and predicted classes are Class W and Class J, respectively. Also, Fig. 10 is the visual explanation where the true and predicted classes are Class J and Class W, respectively. In these figures, the localization maps in the RGB scale are superimposed over the TFR images in the gray scale. However, there is no applicable data when the $(m, 1)$ and $(1, n)$ elements of the confusion matrixes in Fig. 5 are equal to zero. As well as in the correct classification, it can be seen from Figs. 9 and 10 that ResNet18 mainly focuses on the signal intensity pattern below about 10 000 Hz in the incorrect classification. On the other hand, there is a case where ResNet18 captures a different region from a possible important region. For SNR = −20 dB in Fig. 10(a), an important signal intensity pattern should be located between 1500 to 5000 Hz because the true class is Class J0.15. However, the interest region of ResNet18 is concentrated around 10 000 Hz, which may cause the misclassification.



(a) True class: Class W / predicted class: Class J0.15.

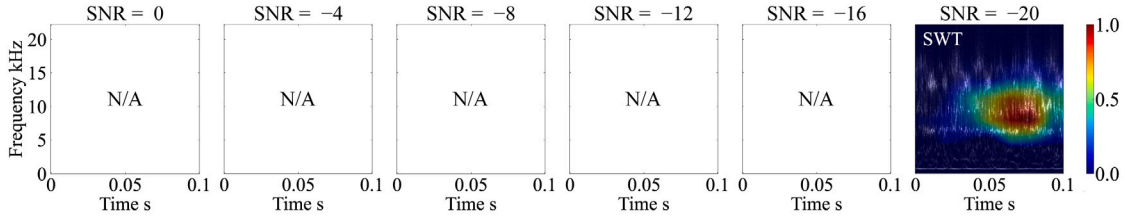


(b) True class: Class W / predicted class: Class J0.17.

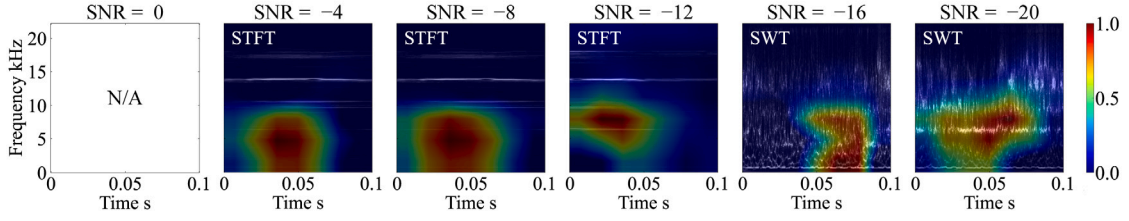


(c) True class: Class W / predicted class: Class J0.20.

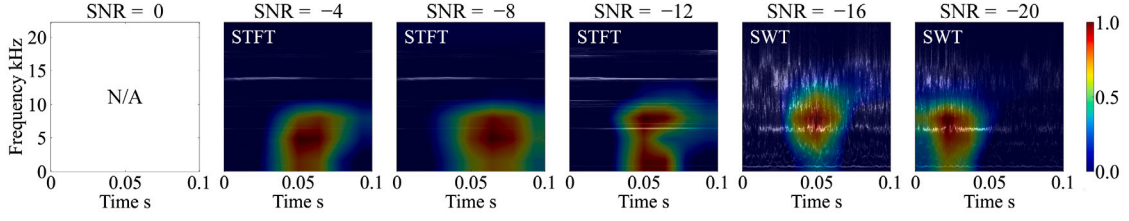
Fig. 9. Visual explanations of incorrect decisions made by ResNet18 for each SNR (true class: Class W/predicted class: Class J). Input images in the gray scale are overlapped with localization maps in the RGB scale obtained by Grad-CAM. Each from left to right shows the visual explanation for SNR = 0, −4, −8, −12, −16 and − 20 dB.



(a) True class: Class J0.15 / predicted class: Class W.



(b) True class: Class J0.17 / predicted class: Class W.



(c) True class: Class J0.20 / predicted class: Class W.

Fig. 10. Visual explanations of incorrect decisions made by ResNet18 for each SNR (true class: Class J/predicted class: Class W). Input images in the gray scale are overlapped with localization maps in the RGB scale obtained by Grad-CAM. Each from left to right shows the visual explanation for SNR = 0, −4, −8, −12, −16 and − 20 dB.

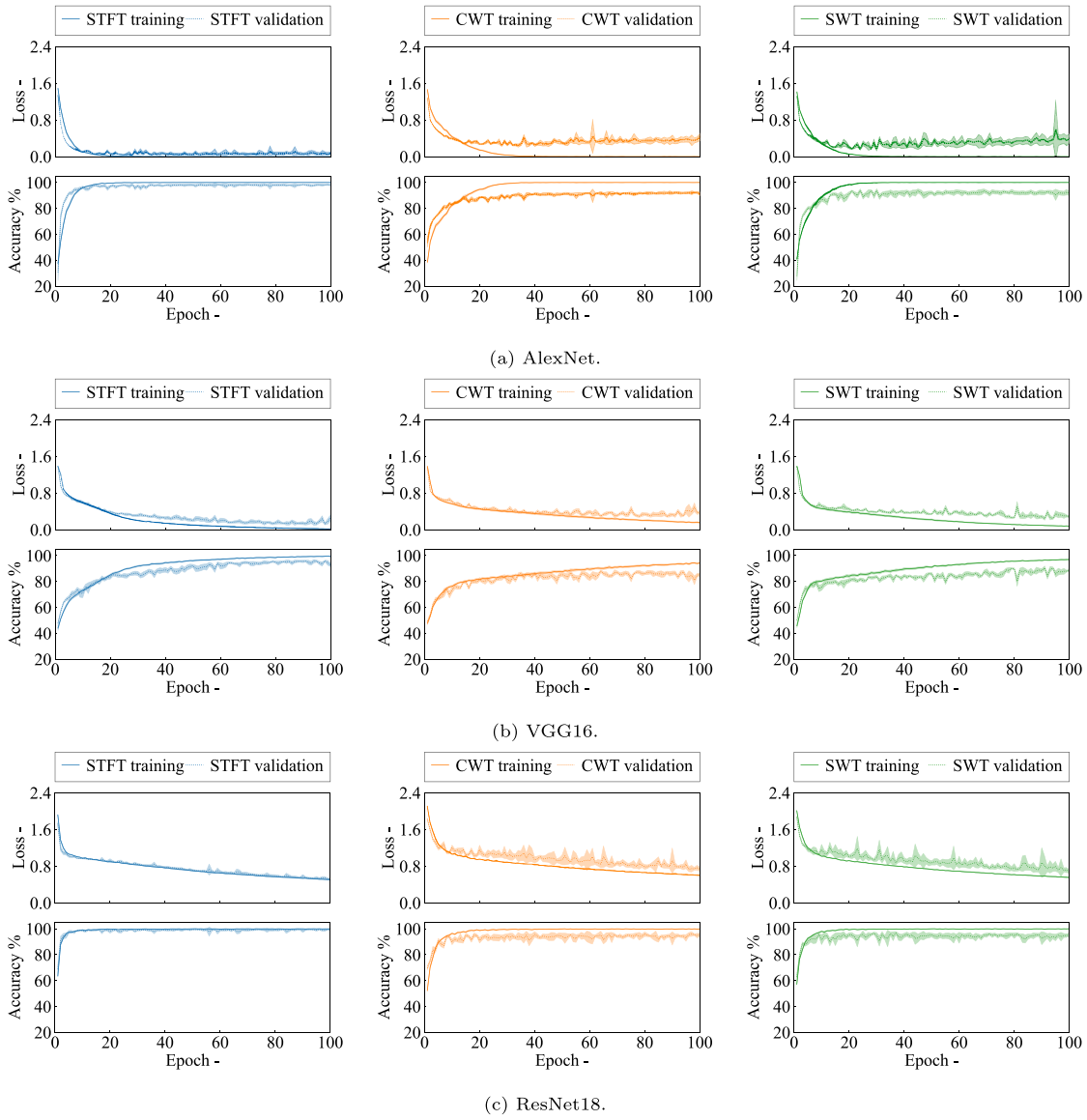


Fig. A.1. History of accuracy and loss in training and validation sessions: (a) AlexNet; (b) VGG16; (c) ResNet18. Solid and dotted lines represent history in training and validation sessions, respectively. Each from left to right illustrates history of STFT (shown in blue), CWT (shown in orange), and SWT (shown in green). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

5. Conclusion

The present study has investigated the CNN-based acoustic identification methods of gas–liquid two-phase jets to detect or analyze anomalies in SFR fast and accurately. First, we have introduced the water flow sound and the gas–liquid jet sound, each of which simulates the background noise and the anomaly sound. Second, as inputs of CNNs, TFRs have been extracted based on time–frequency analysis methods of STFT, CWT, and SWT. Third, as new attempts, we have evaluated the noise resistance of the proposed methods for different SNRs and produced the visual explanations of the decisions of CNNs via Grad-CAM. Our results are summarized as follows:

1. In low-level noise environments, our proposed methods can identify the flow states of gas–liquid two-phase jets with high accuracy. Even in high-level noise environments, the recognition ability of CNNs maintains a certain accuracy, and they can detect gas–liquid two-phase jets.
2. As inputs of CNNs, TFRs based on SWT show their superiority in higher-level noise environments. This is because the scale parameter terms in SWT intensify the signal intensity of the gas–liquid jet sounds in the low-frequency band when the angular frequency increment is fixed.
3. Grad-CAM suggests that the interest regions of ResNet18 are the spectrum peak of the water flow sound and all or part of the signal intensity pattern of the gas–liquid jet sounds. Also, a case can be confirmed where ResNet18 makes an error when its interest region is different from a possible important region.

From a practical point of view, CNN-based acoustic identification methods with Grad-CAM have a merit of applicability to noisy or invisible circumstances including the inside of the SG. This merit leads to the safe operation and fast, accurate, and accountable anomaly detection or analysis in SFR. However, the passive sensing systems in the present study may be unavailable when bubbles are acoustically dominant: sound signals are attenuated significantly through bubbles because of the mismatch of acoustic impedances. Although the present study does not cover this case but covers the early stage of the wastage phenomenon causing the small water/vapor leakage, the active sensing systems should be discussed which measure the attenuation of sound signals.

In future works, we plan to employ deeper CNNs or introduce the noise reduction process because there is room for the improvement in the accuracy of CNNs in high-level noise environments. Also, the shape of heatmaps obtained by Grad-CAM is seven pixels in width and height in this paper. This motivates us to use other CAM methods to increase the resolution of heatmaps. In addition, the background noise involves the sound of boiling and the anomaly sound includes the sound of the sodium-water reaction in actual SFR. Although the present study does not cover these sounds because it is positioned as a fundamental study of proof of concept, we will discuss their physics and consider their effects to the accuracy of CNNs.

CRediT authorship contribution statement

Nao Mikami: Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – Original Draft, Visualization. **Yoshitaka Ueki:** Conceptualization, Methodology, Investigation, Resources, Writing – review & editing, Project administration, Funding acquisition. **Masahiko Shibahara:** Methodology, Resources, Supervision. **Kosuke Aizawa:** Conceptualization, Methodology, Investigation, Resources, Project administration, Funding acquisition. **Kuniaki Ara:** Conceptualization, Methodology, Supervision.

Declaration of competing interest

This manuscript has not been published and is not under consideration for publication elsewhere. We have no conflicts of interest to disclose.

Data availability

Data will be made available on request.

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Appendix. History of accuracy and loss

Fig. A.1 illustrates the history of accuracy and loss for SNR = 0 dB. In this figure, solid and dotted lines represent average accuracy or loss in the training and validation sessions, respectively. It can be seen from Fig. A.1 that accuracy and loss are settled for large epochs in all cases of AlexNet, VGG16, and ResNet18. This result indicates that the hyperparameters in Table 2 are properly adjusted in the validation session.

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